

Predicting Vehicle Maintenance Demand:

Comparison of Logistic
Regression and Random
Forest

Project Showcase

Topic: Supervised Learning, Model Comparison

October 2026



Preview

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Background & Motivation

Logistics fleets rely on the **continuous normal operation of vehicles**. If maintenance needs are not identified in a timely manner, **vehicles may break down during transportation**.

Why does the logistics industry need this project?

Logistics fleets rely on the continuous normal operation of vehicles. If maintenance needs are not identified in a timely manner, vehicles may break down during transportation.

- **Unplanned outage cost:** temporary vehicle shutdown and deployment of replacement vehicles.
- **Emergency maintenance cost:** ad-hoc repairs, towing or rescue fees.
- **Delivery delay cost:** delayed orders and impaired customer experience.
- **Unnecessary maintenance cost:** inspections or maintenance scheduled for vehicles that do not require servicing, consuming manpower, budget and vehicle availability time.

Why this task matters

1

Logistics enterprises need to arrange vehicle maintenance more reasonably, so as to detect maintenance requirements in a timely manner and **reduce unnecessary maintenance**.

2

Using vehicle data to determine maintenance needs can help enterprises schedule maintenance resources, reduce unplanned downtime, and **improve fleet operational efficiency**.

Objectives

REDUCE DOWNTIME AND OPTIMIZE MAINTENANCE COSTS



Compare The Performance Of The Two Models

Based on vehicle records, determine which vehicles may require maintenance.



Analyze Errors In Combination With Operational Requirements

Compare Logistic Regression and Random Forest to see which one yields more accurate predictions.



Establish And Maintain A Demand Forecasting Model

Compare who misses fewer maintenance requirements and who issues fewer invalid alerts to evaluate which model is more suitable for supporting maintenance decisions.

Underreporting: Cases where vehicles require maintenance fail to be identified, which may result in missed maintenance opportunities and raise the risks of malfunctions and out-of-service downtime.



False positive: Alerts are triggered when the vehicle does not require maintenance, which may lead to unnecessary inspections or maintenance schedules.

Dataset Overview

TRAINING AND TEST DATASETS

Dataset	Records	Class 1 share
Training set	56,697	50%
Test set	24,299	50%

Target: Need_Maintenance	
0	No maintenance required
1	Maintenance required

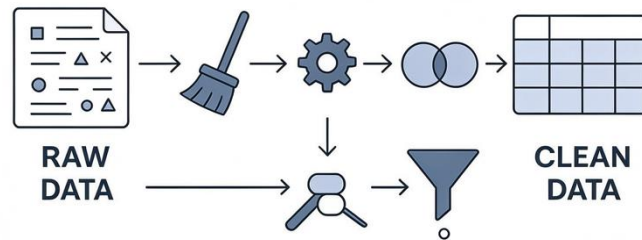
Variable Name	Type	Description
Vehicle_Model	Categorical	Vehicle model
Mileage	Numerical	Distance travelled
Maintenance_History	Categorical	Previous maintenance status
Reported_Issues	Numerical	Number of reported issues
Vehicle_Age	Numerical	Age of the vehicle
Fuel_Type	Categorical	Type of fuel used
Transmission_Type	Categorical	Type of transmission
Engine_Size	Numerical	Engine capacity
From last service	Numerical	Time since the last service
Service_History	Numerical	Number of previous services
Accident_History	Numerical	Number of previous accidents
Fuel_Efficiency	Numerical	Vehicle fuel efficiency
Owner_Type	Categorical	Vehicle ownership category
Tire_Condition	Categorical	Condition of the tires
Intensity	Numerical	Vehicle usage intensity

Data Preparation

CHECKS AND PREPROCESSING

Check	Count
Missing cells (both files)	0
Duplicate rows (within each file)	0
Test rows with identical training inputs	0

Existing split retained. No additional SMOTE.



MODEL INPUTS

Categorical fields

One-hot encoding for both models.

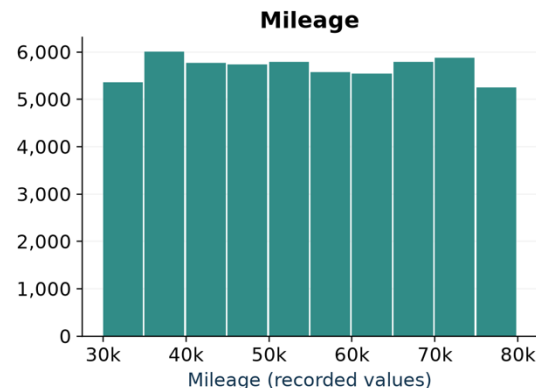
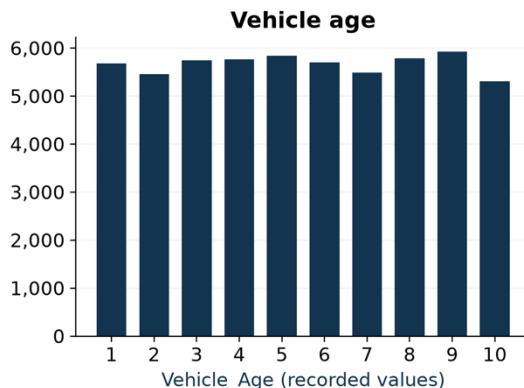
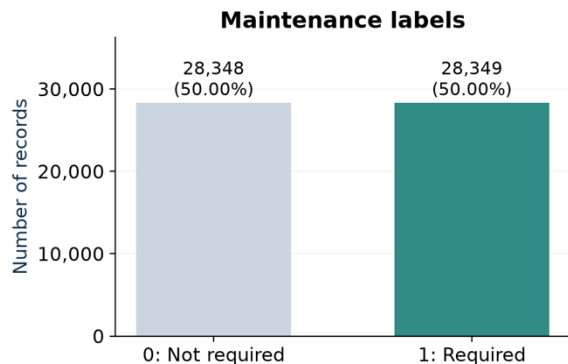
Numerical fields

Standardized for Logistic Regression. Original scale for Random Forest.

Preprocessing is fitted only on the training portion, separately within each CV fold.

EDA: Data Profile

Training data profile | 56,697 records



Key Observations

Balanced maintenance labels

The training set contains 28,348 records without maintenance needs and 28,349 with maintenance needs, giving an approximately 50:50 split.

Broad vehicle-age coverage

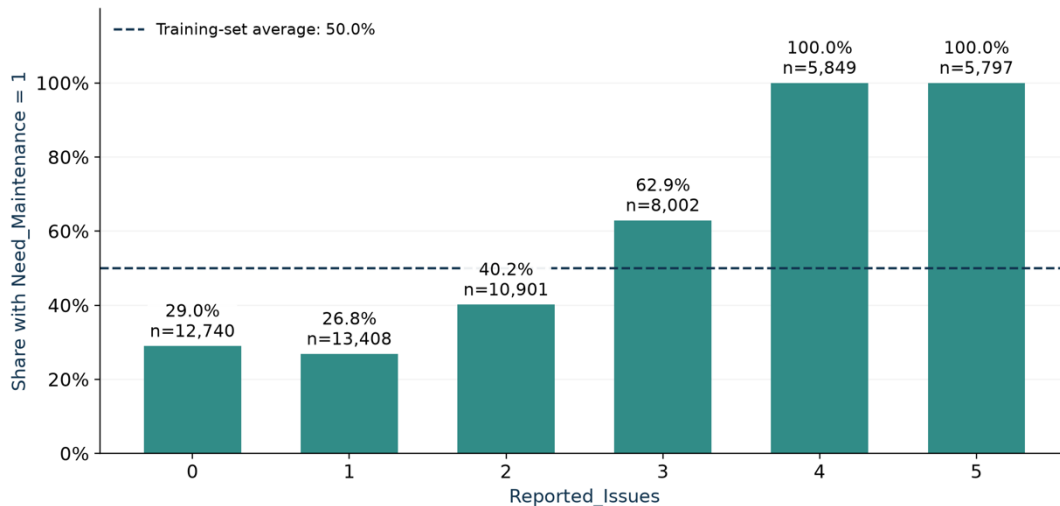
Recorded vehicle ages range from 1 to 10, with similar numbers of records across age groups.

Relatively even mileage distribution

Mileage ranges from 30,003 to 80,000, with a median of 54,707. Records are spread relatively evenly across the displayed intervals.

EDA: Reported Issues

Maintenance need by reported issue count



Implication

Reported issue count shows a strong association with maintenance needs in this dataset and may be useful for identifying records that warrant maintenance review.



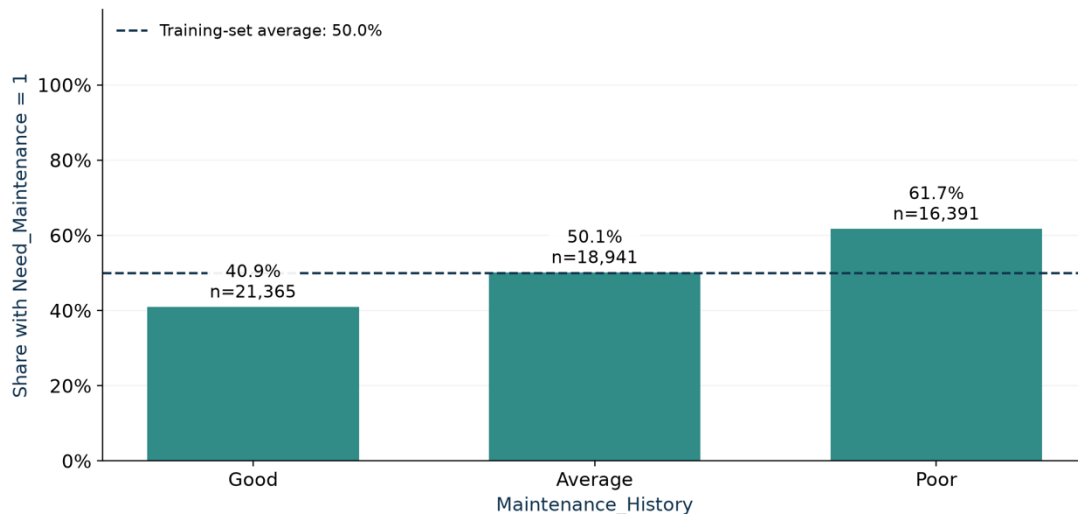
Key Observations

- **0–2 reported issues:** Maintenance need rates remain below the training-set average of 50%.
- **3 reported issues:** The maintenance need rate rises to 62.9%, exceeding the training-set average.
- **4–5 reported issues:** All records in these groups are labelled as requiring maintenance, covering 11,646 records.



EDA: Maintenance History

Maintenance need by maintenance history



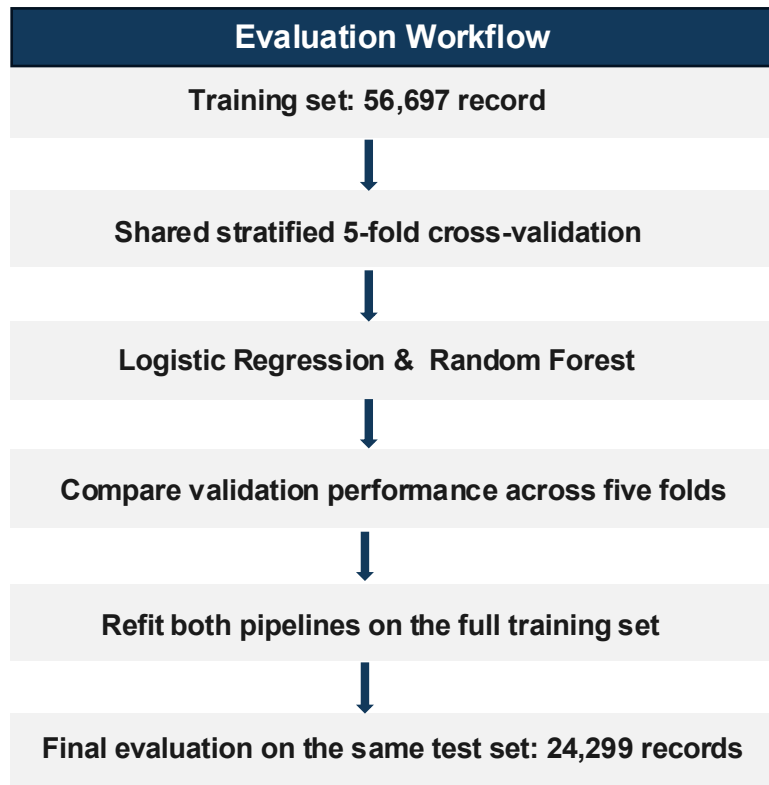
Key Observations

- Good: 40.9% of records require maintenance, below the training-set average of 50%.
- Average: The maintenance need rate is 50.1%, close to the training-set average.
- Poor: 61.7% of records require maintenance, approximately 20.7 percentage points higher than the Good group.

Implication

Maintenance history may help prioritize vehicles for inspection. However, 40.9% of records in the Good group still require maintenance, suggesting that this variable should be considered alongside other vehicle information.

Modeling & Evaluation



Key Settings		
Setting	Logistic Regression	Random Forest
Numerical inputs	Standardized	Original scale
Categorical inputs	One-hot encoding	One-hot encoding
Main parameters	L2 regularization; C = 1.0	300 trees; minimum leaf size = 2
Random seed	42	42
Classification threshold	0.5	0.5

Decision rule: Predict maintenance required when the predicted probability is ≥ 0.5 .

Evaluation control: Fixed model settings; no additional SMOTE and no test-set tuning.

ROC-AUC uses predicted probabilities across thresholds; it is not calculated from the 0.5 decision rule.

Cross-validation Results

Random Forest achieves higher AUC in every fold

Validation fold	Logistic Regression AUC	Random Forest AUC
Fold 1	0.8174	0.8971
Fold 2	0.8173	0.8945
Fold 3	0.8207	0.8947
Fold 4	0.8168	0.8927
Fold 5	0.8190	0.8934
Mean $\pm$ SD	0.8182 $\pm$ 0.0016	0.8945 $\pm$ 0.0017



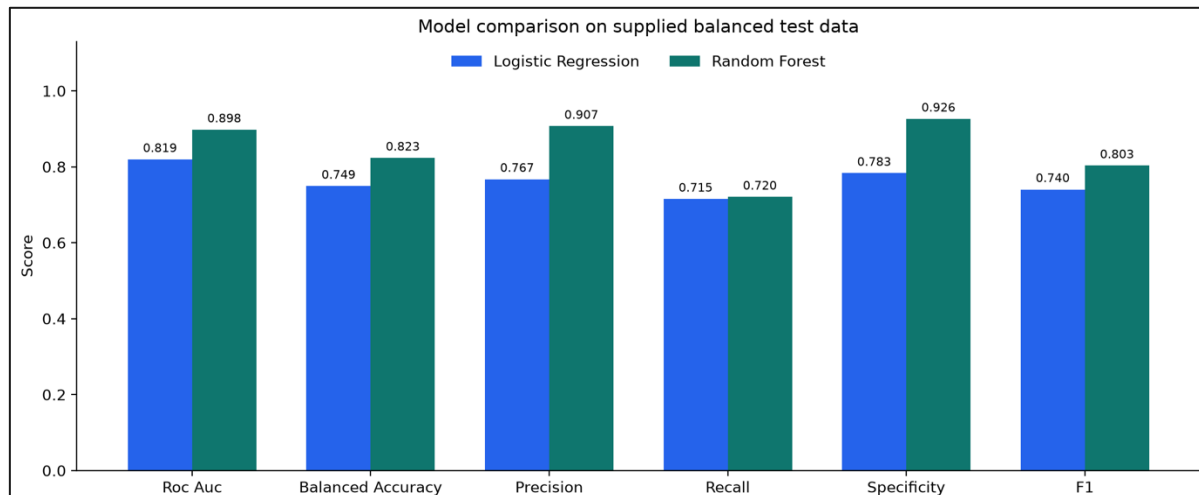
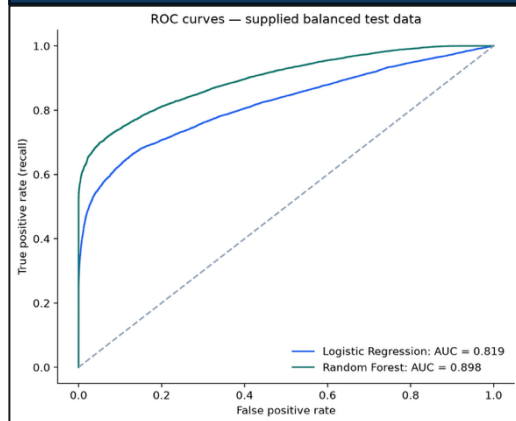
Key Observations

- Random Forest achieves a higher AUC in all five validation folds.
- Its mean AUC exceeds Logistic Regression by **0.0763**.
- Both models show small variation across folds, indicating consistent performance within the supplied training data.

SD = standard deviation across folds. Small variation does not establish performance on independent real-world data.

Test Results

Random Forest achieves 82.32% accuracy and 0.8976 ROC-AUC

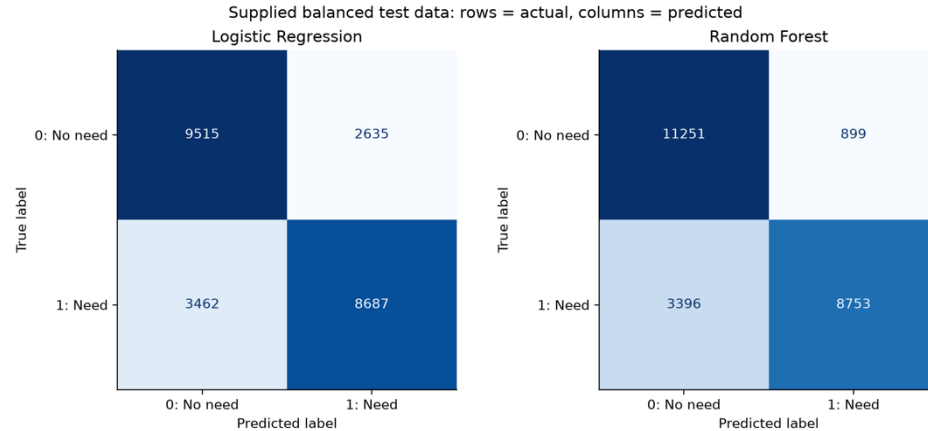


Key Observations

- Random Forest improves accuracy by **7.42 percentage points** and ROC-AUC by **0.0783**.
- Precision improves substantially, while recall increases by only **0.54 percentage points**.
- Test AUC values are close to the cross-validation means for both models.

False Positives & False Negatives

Random Forest produces 1,736 fewer false positives and 66 fewer false negatives.



Error type	Logistic Regression	Random Forest	Reduction
False positives: unnecessary alerts	2,635	899	1,736 (65.9%)
False negatives: missed maintenance needs	3,462	3,396	66 (1.9%)

Operational Interpretation

- The main improvement is fewer alerts for records that do not require maintenance.
- The reduction in missed maintenance needs is limited.
- Random Forest still misses approximately **28.0%** of records that require maintenance at the current threshold.

Feature Importance

Both models rely most on reported issues and service history.

How importance is measured

Shuffle one input variable and measure the decrease in ROC-AUC. A larger decrease indicates greater model reliance on that variable.

Key Observations



- **Reported_Issues** is the most important variable for both models.
- **Service_History** ranks second in both models.
- Random Forest also relies on **Accident_History** and **Engine_Size**.

Interpretation



These rankings explain how the fitted models use the available inputs. They do not establish causal effects or indicate whether a higher value increases maintenance need.

Permutation importance was calculated on the same 2,000 test records, with five shuffles per variable. Error bars show the standard deviation across shuffles.

Key Findings & Next Steps

KEY FINDINGS	• Random Forest performs better overall: 82.32% test accuracy and 0.8976 ROC-AUC. • Its main advantage is fewer false positives: 1,736 fewer unnecessary alerts, compared with only 66 fewer missed needs. • Both models rely most on reported issues and service history.
CURRENT LIMITATIONS	• Results reflect supplied balanced data; real-fleet maintenance prevalence may differ. • Independence between records cannot be confirmed, including possible dependencies from earlier sampling. • This project classifies maintenance needs; future failure timing and actual cost savings were not evaluated.
NEXT STEPS	• Validate on independent operational data, separating vehicles or time periods where appropriate. • Explore decision thresholds using training-side validation, balancing missed needs against false alerts. • Evaluate operational value using inspection capacity, downtime and maintenance costs.



Conclusion: Random Forest is the stronger candidate for further validation in logistics maintenance decision support.